

Basic SQL Queries

SELECT Statement

The SELECT statement is used to retrieve data from one or more tables in a database. It allows you to specify which columns to display and filter data using conditions. This is the most commonly used SQL command for querying information.

Syntax

```
SELECT column1, column2 FROM table_name;
```

For Example:

```
SELECT name, age FROM students;
```

INSERT Statement

The INSERT statement is used to add new rows of data to a table. It allows you to specify values for each column, either partially or completely. This command is essential when you want to populate a database with new records.

Syntax:

```
INSERT INTO table_name (column1, column2) VALUES (value1, value2);
```

For Example:

```
INSERT INTO students (student_id, name, age, major)
```

```
VALUES (101, 'Alice', 22, 'Computer Science');
```

UPDATE Statement

The UPDATE statement is used to modify existing records in a table. You can update one or more columns by specifying conditions to identify which rows to change. Always use the WHERE clause to avoid updating all rows unintentionally.

Syntax:

```
UPDATE table_name SET column1 = value1 WHERE condition;
```

For Example:

```
UPDATE students SET age = 22 WHERE name = 'Ali';
```

DELETE Statement

The DELETE statement is used to remove one or more rows from a table. Similar to UPDATE, it is crucial to use the WHERE clause to specify which records to delete, as omitting it will delete all rows in the table.

Syntax:

```
DELETE FROM table_name WHERE condition;
```

For Example:

```
DELETE FROM students WHERE age = 22;
```

WHERE Clause

The WHERE clause is used to filter records based on specified conditions. It can be combined with SELECT, UPDATE, and DELETE statements to narrow down the affected rows.

Example:

```
SELECT * FROM students WHERE age > 20;
```

ORDER BY Clause

The ORDER BY clause is used to sort the result set in ascending (default) or descending order based on one or more columns. It enhances the presentation of query results by organizing the output.

Example:

```
SELECT * FROM students ORDER BY age DESC;
```

LIMIT Clause

The LIMIT clause is used to restrict the number of rows returned by a query. It is useful when you only need to retrieve a subset of records, improving performance and efficiency.

Example:

```
SELECT * FROM students LIMIT 5;
```

DISTINCT Keyword

The DISTINCT keyword is used to return unique values from a specified column, eliminating duplicate entries in the query result.

Example:

```
SELECT DISTINCT age FROM students;
```

Aggregate Functions

Aggregate functions perform calculations on a set of rows and return a single value. Common functions include COUNT(), AVG(), SUM(), MIN(), and MAX().

Examples:

Count total rows:

```
SELECT COUNT(*) FROM students;
```

Find the average age:

```
SELECT AVG(age) FROM students;
```

Find the maximum age:

```
SELECT MAX(age) FROM students;
```